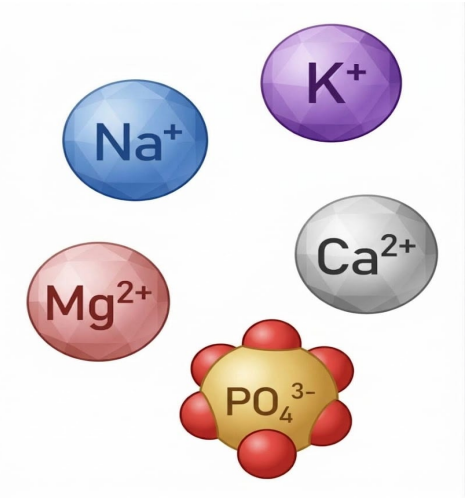
Hawler Medical University
College of Medicine

Electrolyte Balance

Biochemistry Lab 6
Second Stage (2026–2027)



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Learning Outcomes

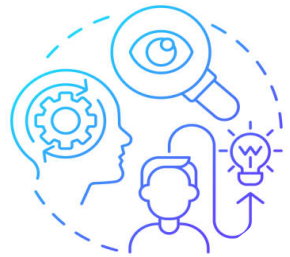
Upon completion, students will be able to:

✓Define Core Concepts: Differentiate between cations (+), anions (-), and clinical measurement units (mEq/L vs. mmol/L).

✓ Outline Systemic Roles: Describe electrolyte regulation of fluid balance, neural transmission, and muscle function.

✓ Analyze Clinical Profiles: Evaluate the etiologies and signs of serum imbalances (Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, HCO_3^-, PO_4^{3-}).

✓ Explain Assay Principles: Describe the spectrophotometric reaction between serum calcium and o-CPC at alkaline pH.



Core Concepts & Charge Balance

- Electrolytes are charged solutes dissolved in body water that maintain cellular function and homeostatic balance.
- Clinical units measure total concentration in mmol/L and electrical charge combining power in mEq/L.
- Electrolytes divide into positively charged cations and negatively charged anions.
- Electrical Neutrality: Total Cations (+) = Total Anions (-) in every fluid space.

Major Electrolytes in human body

Cations (+)

- Sodium: Na^+ (ECF)
- Potassium: K^+ (ICF)
- Calcium: Ca^{2+} (ECF)
- Magnesium: Mg^{2+} (ICF)

Anions (-)

- Chloride: Cl^- (ECF)
- Bicarbonate: HCO_3^- (ECF)
- Phosphate: PO_4^{3-}(ICF)

Primary Physiological Roles of Electrolytes

Electrolytes are vital minerals with an electrical charge that keep your body functioning through five primary roles:

1) Regulate water distribution
2) Muscle contraction
3) Nerve impulse transmission
4) Blood clotting
5) Regulate enzyme reactions

Sodium (Na^+): Clinical Overview

- Primary Role: Major extracellular cation regulating osmolality, ECF volume, and acid–base balance.

- Reference Interval: 135–145 mEq/L (varies slightly by laboratory).

- Function: Drives neuromuscular excitability (Na^+/K^+-ATPase) and fluid distribution.

Hyponatremia (< 135 mEq/L)

- Etiology: Excess water retention or net sodium depletion.
- GI Losses: Vomiting, diarrhea.
- Dilutional: Heart failure (CHF), renal failure, Syndrome of Inappropriate Antidiuretic Hormone (SIADH).
- Endocrine: Addison's disease (hypoaldosteronism causing renal Na^+ wasting).

Hypernatremia (> 145mEq/L)

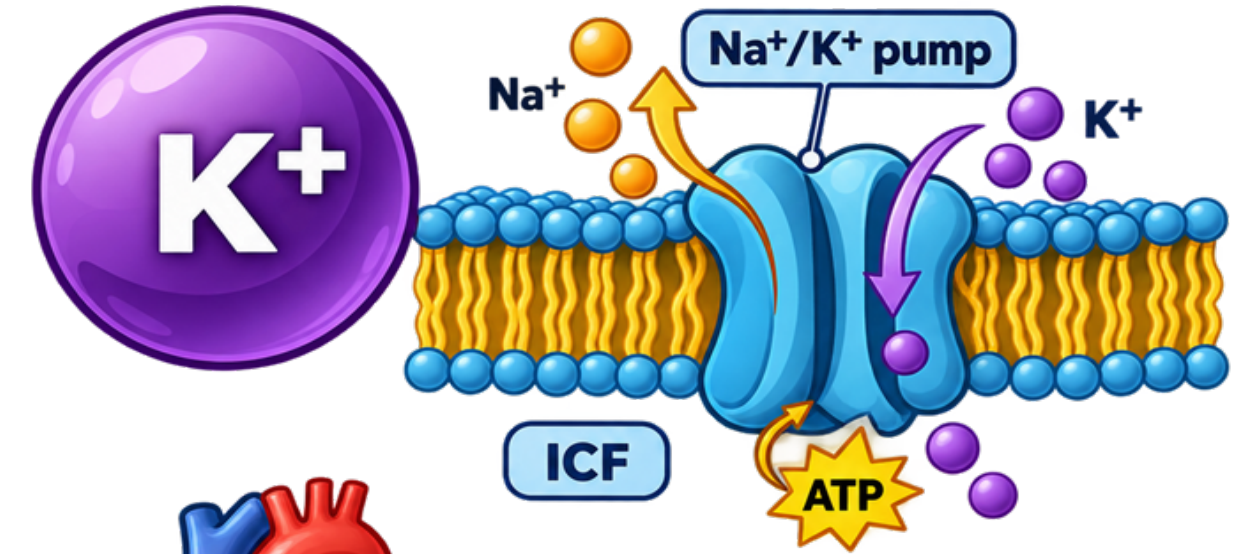
- Etiology: Unreplaced water loss or sodium overload.
- Fluid Loss: High fever, severe dehydration, decreased ADH activity (Diabetes Insipidus).
- Exogenous / Hormonal: Excessive salt intake, Primary Hyperaldosteronism (Conn syndrome), or Cushing syndrome (mineralocorticoid excess).

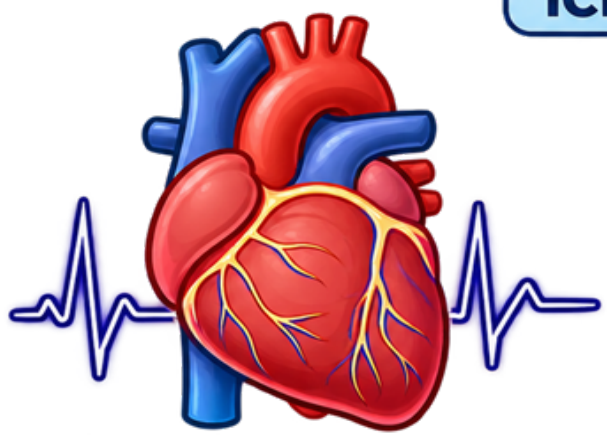
Potassium (K^+): Clinical Overview

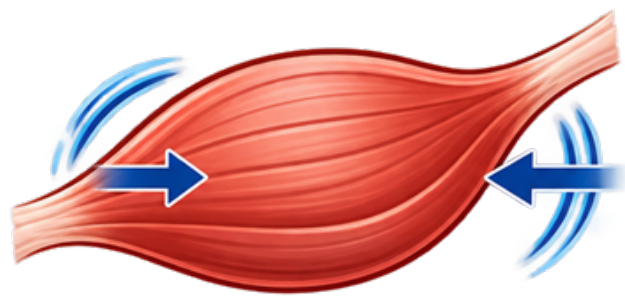
- Potassium: Principal ICF cation (3.5–5.0 mEq/L) maintaining cell volume, membrane potential, and enzyme function.
- Extracellular: Critical for cardiac/skeletal muscle contraction and nerve impulse conduction.
- Intracellular: Regulates enzymatic activity, intracellular osmolality, and acid–base status.

❖Systemic Regulation: Maintained by renal excretion, hormonal axes (Aldosterone/Insulin), and Na^+/K^+-ATPase pumps.

Potassium K+







Hypokalemia (< 3.5 mEq/L)

- Etiology: Decreased intake, increased excretion, or transcellular shifting into cells.
- GI Losses: Prolonged vomiting, diarrhea.
- Renal Wasting: Diuretic therapy, metabolic alkalosis, Cushing's disease (excess mineralocorticoids).

❖Clinical Features: Muscle weakness, cramps, constipation, and cardiac arrhythmias.

Hyperkalemia (> 5.0 mEq/L)

- Etiology: Reduced renal excretion, excessive intake (IV therapy), or cellular release.
- Transcellular Shifts: Acidemia, relative insulin deficiency, or cell injury/rhabdomyolysis.
- Renal & Endocrine: Renal failure, glomerulonephritis, and Addison's disease (hypoaldosteronism causing K+ retention).

❖Clinical Significance: High risk for lethal cardiac conduction delays and arrhythmias.

Calcium (Ca^{2+}): Clinical Overview

- Calcium (Ca^{2+}): Most abundant body mineral (1.0–1.4 kg; 99% in bones/teeth) with a standard reference interval of 4.5–5.0 mEq/L (8.8–10.4 mg/dL).

- Plasma fractions circulate in three forms:

free/ionized (biologically active).

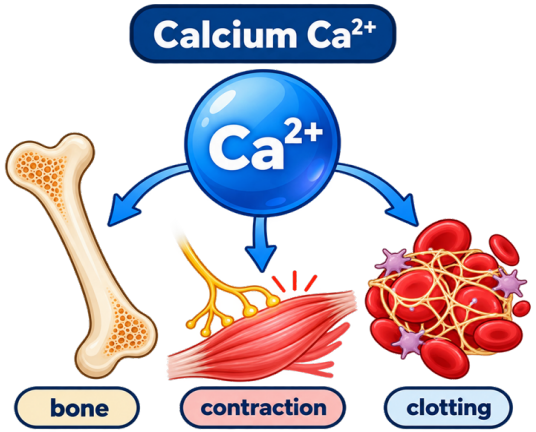
protein-bound (albumin).

complexed (citrate/phosphate/bicarbonate).

Calcium (Ca^{2+}): Clinical Overview

- Biological Functions: Essential for neuromuscular excitation, skeletal/cardiac muscle contraction, coagulation cascade, neural transmission, and Vitamin B12 absorption.

- Regulation: Endocrine control by Parathyroid Hormone (PTH), Vitamin D3 (calcitriol), and Calcitonin.



Hypocalcemia (< 4.3 mEq/L or < 8.8 mg/dL)

- Decreased Intake/Absorption: Vitamin D3 deficiency, malabsorption syndromes.
- Hypoalbuminemia: Malnutrition, severe burns (decreases bound fraction; free (Ca^{2+}) remains normal).
- Endocrine/Bone Mechanisms: Hypoparathyroidism, elevated calcitonin, or post-parathyroidectomy "hungry bone syndrome.“

❖Clinical Features: Neuromuscular irritability, paresthesias, cramps, tetany, seizures, and QT prolongation.

Hypercalcemia (> 5.3 mEq/L or > 10.4 mg/dL)

- Bone Resorption: Primary hyperparathyroidism, osteolytic malignancies.
- Excess Intake/Absorption: Vitamin D3 toxicity, hyperalbuminemia (pseudohypercalcemia via severe dehydration).
- Renal Retention: Thiazide diuretics, Addison's disease.

❖ Clinical Manifestations: Fatigue, depression, anorexia, muscle weakness, nephrolithiasis (renal calculi), polyuria, polydipsia, and cardiac conduction alterations.

Magnesium (Mg^{2+}): Clinical Overview

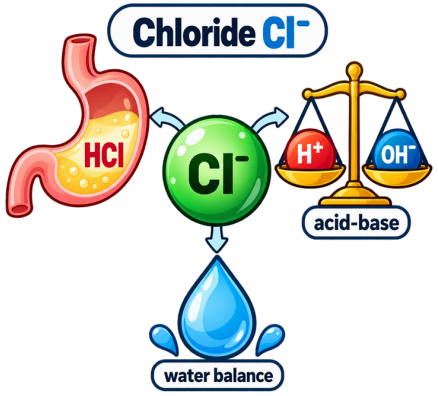
- Magnesium (Mg^{2+}) [1.5–2.5 mEq/L (1.8–3.0 mg/dL)]: Essential cofactor for ATP synthesis, neuromuscular transmission, and cardiac rhythm regulation.
- Hypomagnesemia (< 1.5 mEq/L): Caused by alcoholism or diuretics; leads to tetany and arrhythmias.
- Hypermagnesemia (> 2.5 mEq/L): Caused by renal failure; leads to hyporeflexia, bradycardia, and hypotension.

Chloride (Cl^-): Clinical Overview

➢Chloride (Cl^-) : Major ECF anion (95–105 mEq/L) maintaining electroneutrality, osmotic pressure, and fluid balance alongside Na^+.

➢Function:

- Essential for acid–base regulation and fluid balance.
- Contributes to gastric juice production as hydrochloric acid (HCl).
- Regulated primarily via renal reabsorption and excretion.



Hypochloremia (< 95 mEq/L)

- Etiology: Loss of gastric contents, renal wasting, or dilution.
- GI Losses: Prolonged vomiting or gastric suctioning (loss of HCl leading to metabolic alkalosis).
- Renal Losses: Salt-wasting nephropathy or loop/thiazide diuretic therapy.

Hyperchloremia (> 105 mEq/L)

- Etiology: Hemoconcentration, excessive chloride loading, or renal retention.
- Fluid Status: Severe dehydration.
- Renal/Exogenous: Excessive saline (0.9% NaCl) infusion, renal tubular acidosis, or decreased renal excretion.
- Clinical Note: Cl^- levels must always be interpreted alongside Na^+, HCO_3^-,and acid–base status to evaluate anion gap disorders.

Bicarbonate (HCO_3^-): Clinical Overview

- Bicarbonate (HCO_3^-) [22–28 mEq/L]: Major extracellular anion and primary chemical buffer critical for maintaining physiological blood pH and carbon dioxide transport.

- Hypobicarbonatemia (< 22 mEq/L): Caused by diabetic ketoacidosis or severe diarrhea; leads to metabolic acidosis and compensatory hyperventilation.

- Hyperbicarbonatemia (> 28 mEq/L): Caused by prolonged vomiting or diuretic therapy; leads to metabolic alkalosis and neuromuscular excitability.

Phosphate (PO_4^{3-}): Clinical Overview

- Phosphate (PO_4^{3-}) [2.5–4.5 mg/dL]: Principal intracellular anion critical for cellular ATP synthesis, energy trapping, and skeletal bone mineralization..

- Hypophosphatemia (< 2.5 mg/dL): Caused by hyperparathyroidism; leads to ATP depletion and muscle weakness.
- Hyperphosphatemia (> 4.5 mg/dL): Caused by chronic kidney disease; leads to soft-tissue calcification and tetany.

Principle (o-CPC Method)

Calcium ions (Ca^{2+}) in serum react with o-cresolphthalein complexone (o-CPC) in an alkaline medium (buffered usually around pH 10.5 with 2-amino-2-methyl-1-propanol) to form a **violet-colored complex**.

$$Ca^{2+} + \text{o-CPC} \xrightarrow{\textbf{Alkaline pH}} \text{Calcium-o-CPC complex } \textbf{(Violet)}$$

❖Pre-Analytical Errors: Avoid hemolysis, tourniquet stasis (>1 min), and EDTA/oxalate tubes to prevent inaccurate results.

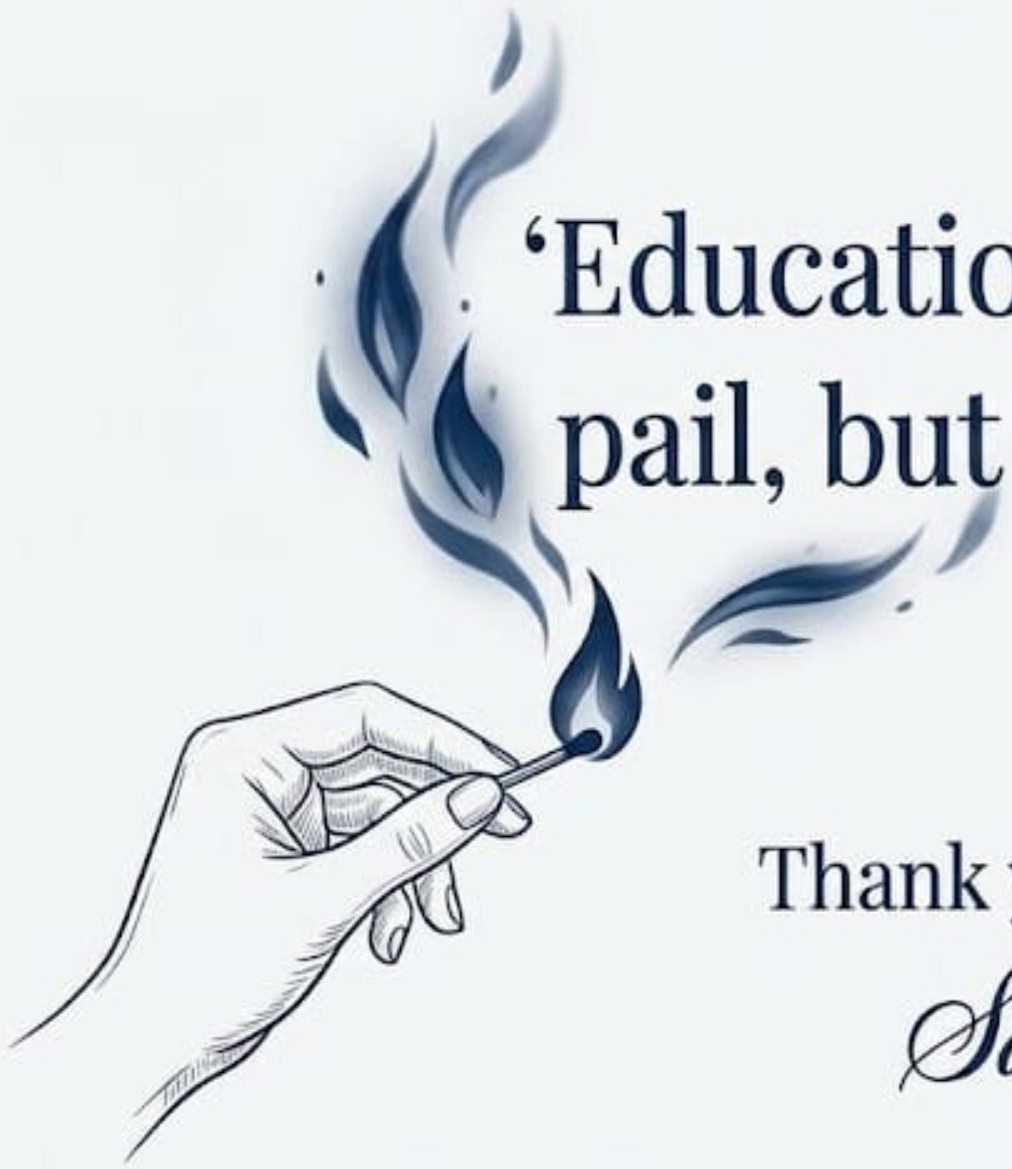
Procedure

- Combine 1.0 mL Working Reagent and 20 µL Serum/Plasma.
- Incubation: Mix thoroughly and incubate for 5 minutes at room temperature.
- Measurement: Read the absorbance at 578 nm

$$\text{Serum Calcium (mg/dL)} = (A_{test} \div A_{std}) \times C_{std}$$

References

- Tietz Fundamentals of Clinical Chemistry and Molecular Diagnostics. Elsevier.
- Henry’s Clinical Diagnosis and Management by Laboratory Methods. Elsevier.
- Guyton and Hall Textbook of Medical Physiology. Elsevier.
- KDIGO clinical resources for kidney function and electrolyte disorders.



‘Education is not the filling of a pail, but the lighting of a fire.’

— W.B. Yeats

Thank you for keeping the flame alive.

Sarween Ahmad Rasool